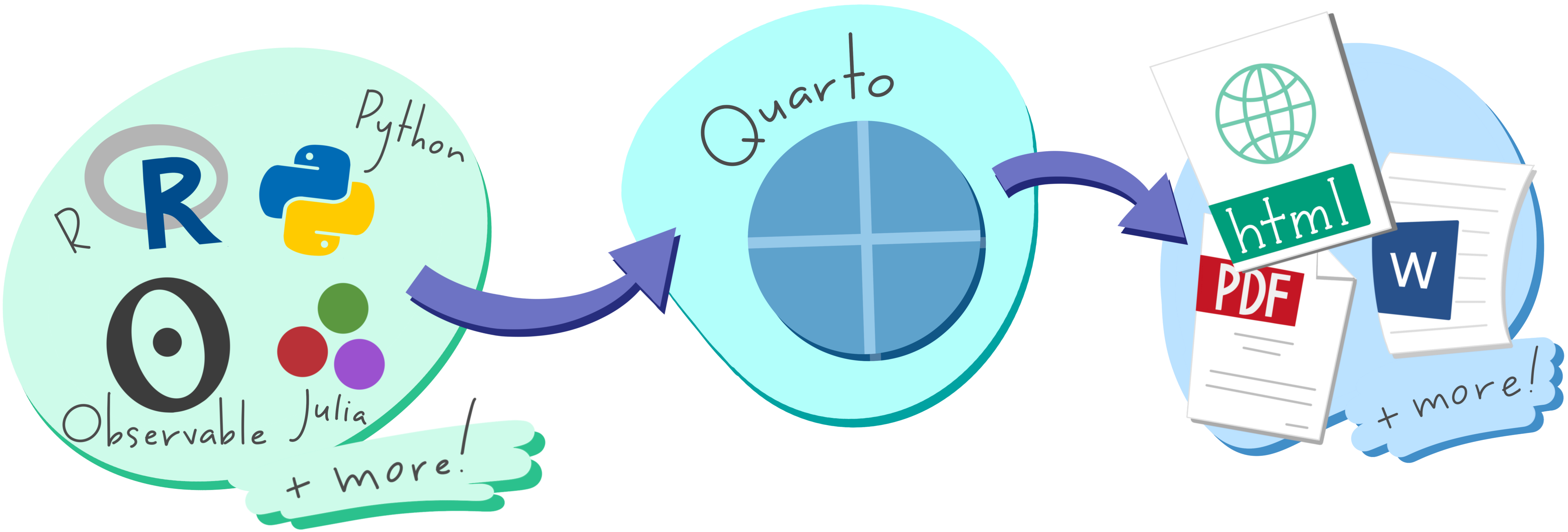
Dynamic documents in R

reproducible research with Quarto

2026-07-30



Artwork from “Hello, Quarto” keynote by Julia Lowndes and Mine Çetinkaya-Rundel, presented at RStudio Conference 2022. Illustrated by Allison Horst.

````
---
title: "ggplot2 demo"
author: "Norah Jones"
date: "5/22/2021"
format:
  html:
    fig-width: 8
    fig-height: 4
    code-fold: true
---

## Air Quality

@fig-airquality further explores the impact of
temperature on ozone level.

```{r}
#| label: fig-airquality
#| fig-cap: Temperature and ozone level.
#| warning: false

library(ggplot2)

ggplot(airquality, aes(Temp, Ozone)) +
  geom_point() +
  geom_smooth(method = "loess"
)
```
````

Source

ggplot2 demo

Norah Jones

May 22nd, 2021

Output

Air Quality

Figure 1 further explores the impact of temperature on ozone level.

▶ Code

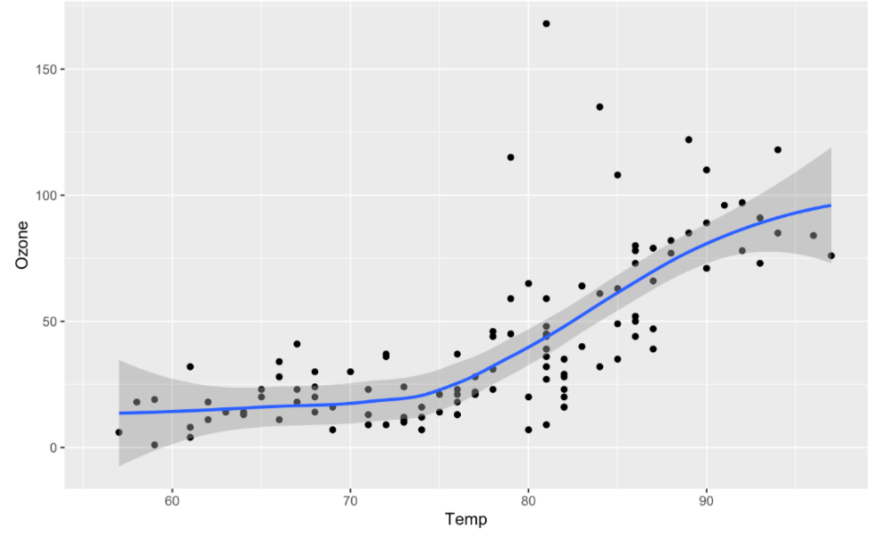


Figure 1: Temperature and ozone level.

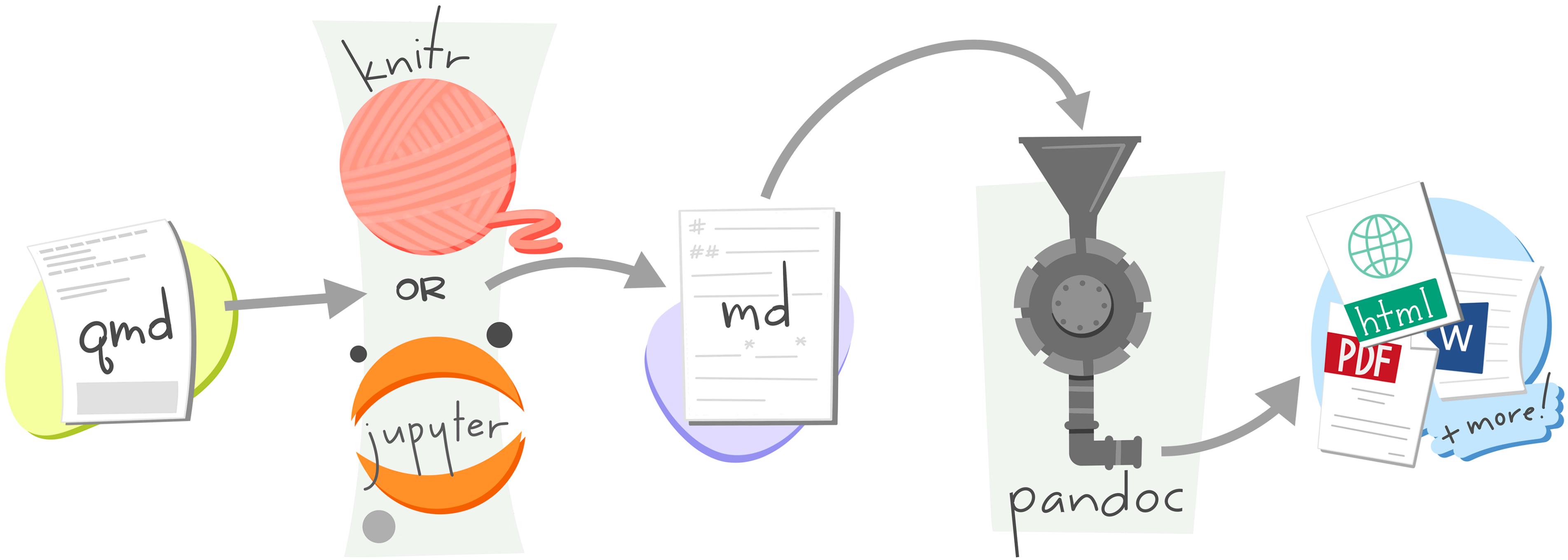
Quarto

Dynamic: code and text in same document

Reproducible: re-run your analysis, re-render your document

Flexible: output to different formats easily

Rendering



Your Turn 1

Create a new Quarto file. Go to File > New File > Quarto Document. Press OK. Save the file and press the “Render” button.

```
---
title: "Untitled"
format: html
---

## Quarto

Quarto enables you to weave together content and executable code into a finished document.
To learn more about Quarto see <https://quarto.org>.

## Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code.
You can embed code like this:

```{r}
1 + 1
```

You can add options to executable code like this

```{r}
#| echo: false
2 * 2
```

The `echo: false` option disables the printing of code (only output is displayed).

```

document metadata

plain text

code chunks

# Quarto

## Prose

## Code

## Metadata

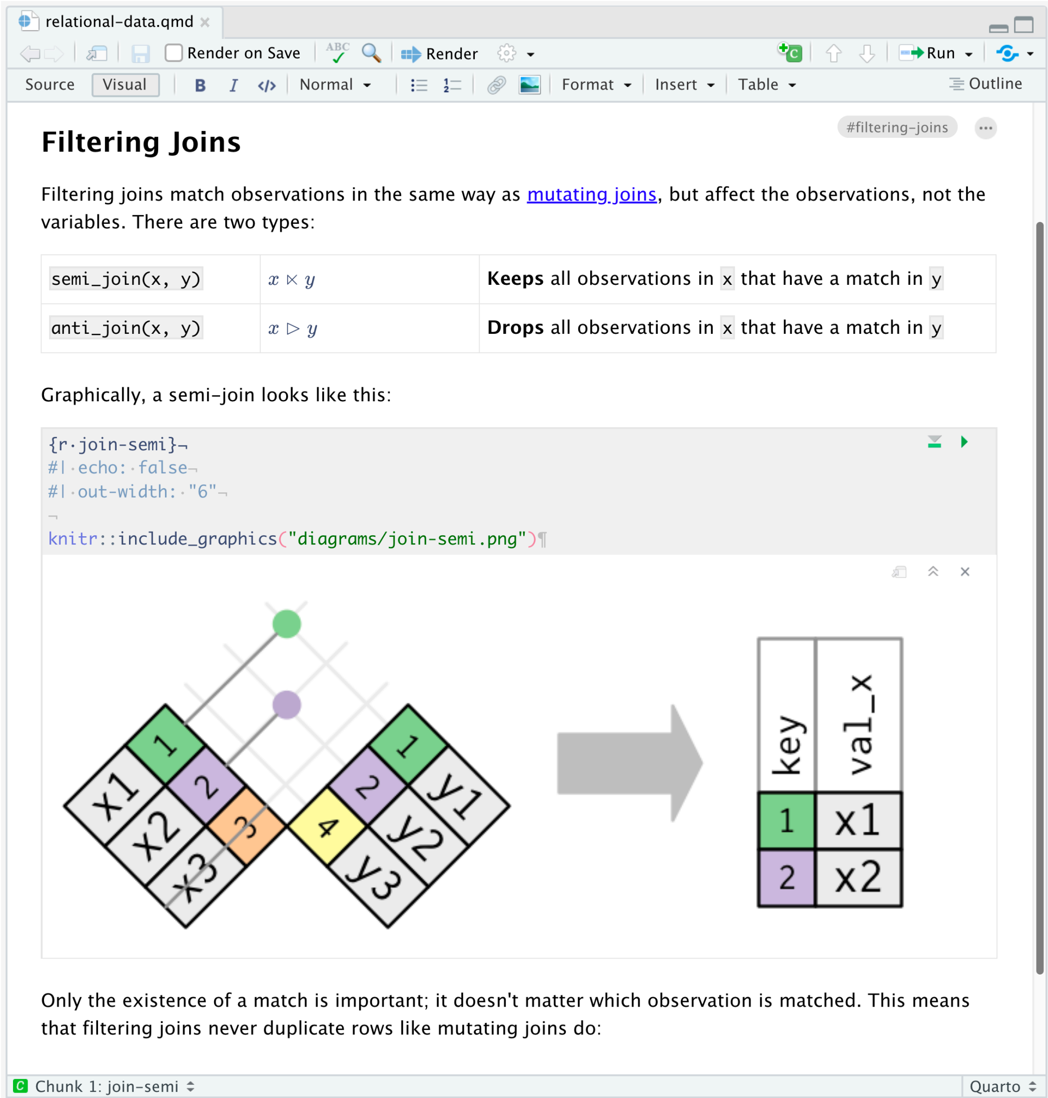
# Quarto

## Prose = *Markdown*

## Code

## Metadata

# the Visual Editor



# Basic Markdown Syntax

```
italic   **bold**

italic   __bold__
```

Basic Markdown Syntax

```
# Header 1

## Header 2

### Header 3
```

Basic Markdown Syntax

```
<http://example.com>

[linked phrase](http://example.com)
```

Learn more about Markdown Syntax with the ten-twenty minute tutorial on markdown at *https://commonmark.org/help/tutorial*.

Your Turn 2 (open `exercises.qmd`)

Read this short introduction to the Visual Editor: https://quarto.org/docs/visual-editor/

Use the Visual Editor to stylize the text from the Gapminder website **below. Experiment with bolding, italicizing, making lists, etc.**

Quarto

Prose

Code = *R code chunks*

Metadata

Code chunks

````
```{r}
#| label: howl
#| echo: false
my_generation |>
  select(best_minds) |>
  filter(destroyed == "by madness")
```
````

Code chunks

code engine

code fences
(three backticks)

````
```{r}
#| label: howl
#| echo: false
my_generation |>
  select(best_minds) |>
  filter(destroyed == "by madness")
```
````

Code chunks

chunk options

```
```{r}
#| label: howl
#| echo: false
my_generation |>
  select(best_minds) |>
  filter(destroyed == "by madness")
```
```

option values

Code chunks

code chunk comments

````
```{r}
#| label: howl
#| echo: false
my_generation |>
  select(best_minds) |>
  filter(destroyed == "by madness")
```
````

Chunk options

Option	Effect
`include: false`	run the code but don't print it or results
`eval: false`	don't evaluate the code
`echo: false`	run the code and output but don't print code
`message: false`	don't print messages (e.g. from a function)
`warning: false`	don't print warnings
`fig.cap: "Figure 1"`	caption output plot with "Figure 1"

See the Quarto documentation

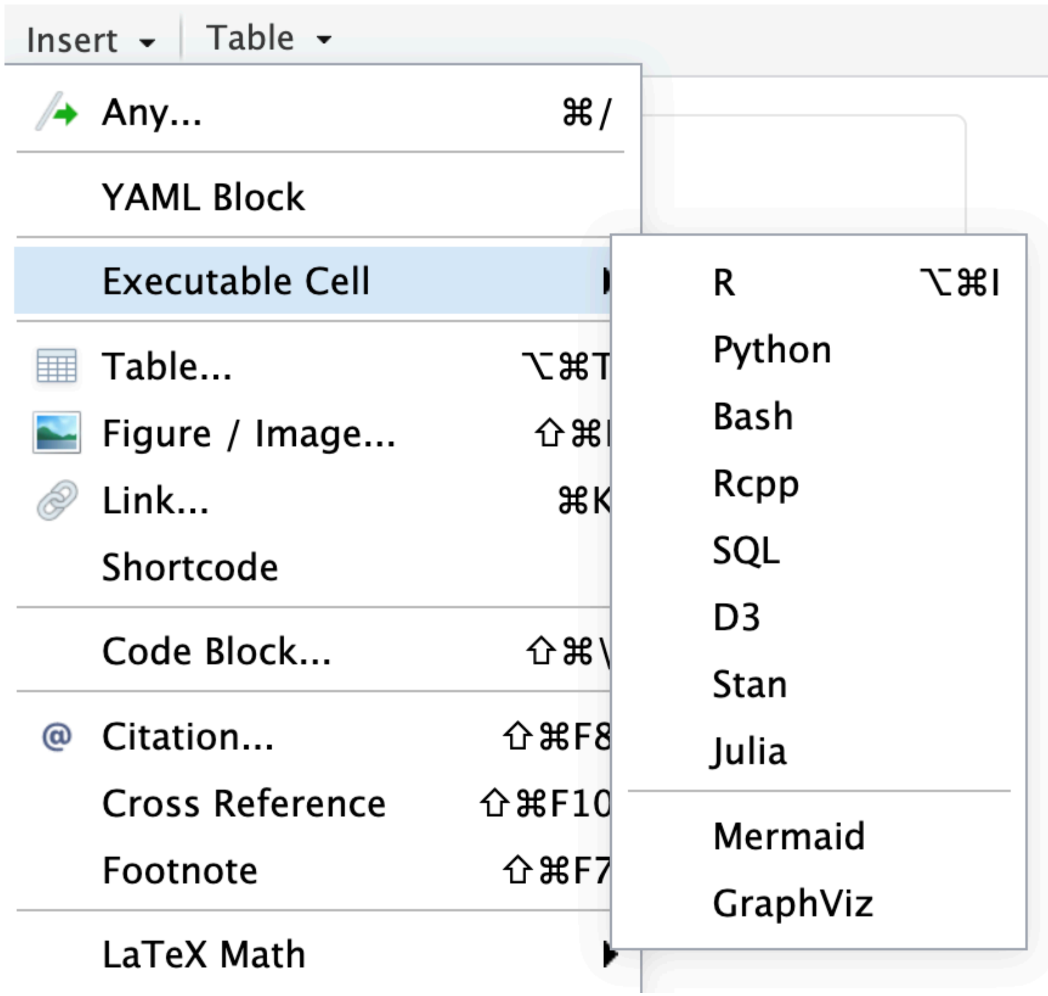
Other languages

First-class support for Python, Julia, and Observable JS

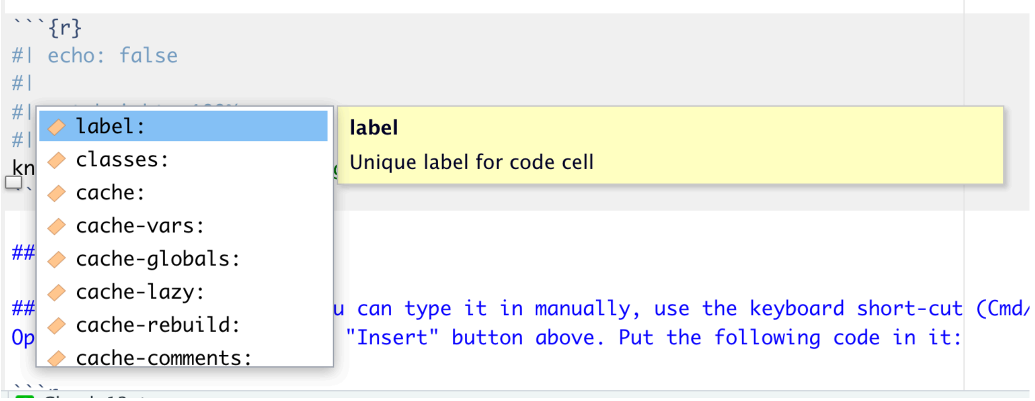
Supports Jupyter notebooks

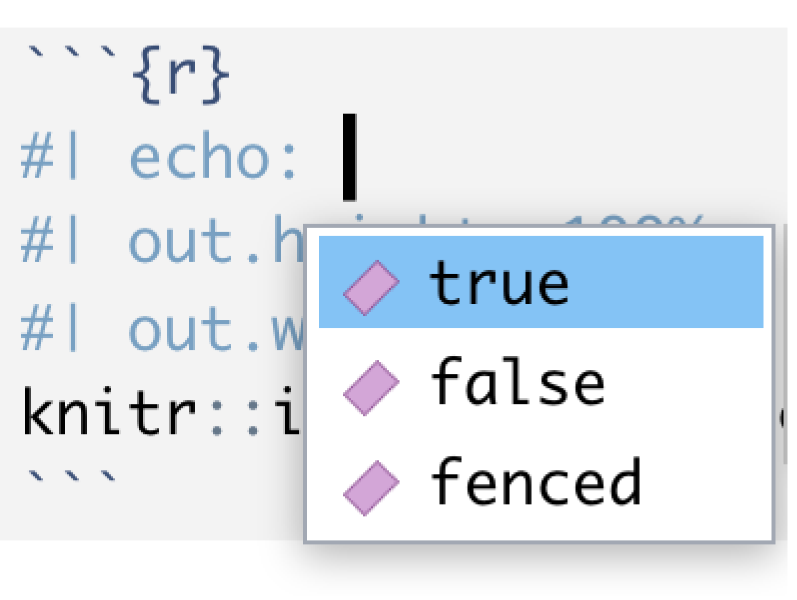
57 knitr engines

Insert code chunks with `cmd/ctrl + alt/option + I` or Visual Editor



Press `tab` to autocomplete chunk options





Your Turn 3

Create a code chunk. You can type it in manually, use the keyboard short-cut (Cmd/Ctrl + Option/Alt + I), or use the “Insert” button above. Put the following code in it:

```
gapminder |>
  slice(1:5) |>
   gt()
```

Render the document

Your Turn 4

Add `echo: false` to the code chunk you created and re-render. What's the difference in the output?

Inline Code

```
who wept at the romance of the
streets with their pushcarts full of
onions and bad music,
who sat in boxes breathing in the
darkness under the bridge, and rose
up to build `r select(instruments,
harpsichord)` in their lofts,
who coughed on the `r length(floors)`
floor of Harlem crowned with flame
under the tubercular sky surrounded
by orange crates of theology,
who scribbled all night rocking and
rolling over lofty incantations which
in the yellow morning were stanzas of
gibberish,
```

Inline Code

```
who wept at the romance of the
streets with their pushcarts full of
onions and bad music,
who sat in boxes breathing in the
darkness under the bridge, and rose
up to build `r select(instruments,
harpsichord)` in their lofts,
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floor of Harlem crowned with flame
under the tubercular sky surrounded
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who scribbled all night rocking and
rolling over lofty incantations which
in the yellow morning were stanzas of
gibberish,
```

single backtick + r

any R code

Your Turn 5

Remove `eval: false` so that Quarto evaluates the code.

Use `summarize()` and `n_distinct()` to get the the number of unique years in gapminder and save the results as `n_years`.

Use inline code to describe the data set in the text below the code chunk and re-render.

Quarto

Prose

Code

Metadata = *YAML*

YAML Metadata

```yaml
---
title: "Howl"
author: "Allen Ginsberg"
date: "March 18, 1956"
format:
  html: default
  pdf:
    toc: true
    number-sections: true
---
```

Output formats

Format	Outputs
`html`	HTML
`pdf`	PDF
`word`	Word `.docx`
`odt`	OpenOffice `.odt`
`gfm`	GitHub-flavored Markdown
`revealjs`	Reveal Slides (HTML)
`beamer`	Beamer Slides (PDF)
`pptx`	Powerpoint Slides

Your Turn 6

Set figure chunk options to the code chunk below, such as `dpi`, `fig.width`, and `fig.height`. Run `knitr::opts_chunk$get()` in the console to see the defaults.

Add your name to the YAML header using `author: Your Name`.

Change `format: html` to use the `toc: true` and `code-fold: true` options and re-render

Parameters

```
---
params:
  param1: x
  param2: y
  data: df
---
```

Calling parameters in R

```
params$param1
params$param2
params$data
```

From the Console

```
quarto::quarto_render(
  "document.qmd",
  execute_params = list(param1 = 0.2, param2 = 0.3)
)
```

Your Turn 7

Change the `params` option in the YAML header to use a different continent. Re-render.

```r
gapminder |>
  filter(continent == params$continent) |>
  ggplot(aes(x = year, y = lifeExp, group = country, color = country)) +
  geom_line(lwd = 1, show.legend = FALSE) +
  scale_color_manual(values = country_colors) +
  theme_minimal(14) +
  theme(strip.text = element_text(size = rel(1.1))) +
  ggtitle(paste("Continent:", params$continent))
```

Bibliographies and citations

Bibliography files: `.bib`, Zotero, others

Citation styles: `.csl`

`[@citation-label]`

Visual Editor's citation wizard can help!

Including bibliography files in YAML

```yaml
---
bibliography: file.bib
csl: file.csl
---
```

the Visual Editor can also manage this for you.

Your turn 8

Cite Causal Inference in text below. Using the citation wizard, find the right citation under My sources > Bibliography.

Add the American Journal of Epidemiology CSL to the YAML using csl: `aje.csl`

Re-render

Make cool stuff in Quarto!

- Books
- Blogs
- These slides!

See the Gallery for inspiration

Resources

Quarto Documentation: **A comprehensive but friendly introduction to Quarto. Written in Quarto!**

R for Data Science: **A comprehensive but friendly introduction to the tidyverse. Free online. Written in Quarto!**

Posit Recipes: **Common code patterns in R (with some comparisons to SAS)**